

Spreading 40 Points in the Plane

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Abstract

We place 40 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 37.08126+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 41-point configuration gives 38.70713; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 40 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 37.08126\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 40$ is not on the page. Deleting a point from our 41-point configuration gives 38.70713; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n40.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
128308884144257231989853
denominator:
3460207612448793335170
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

40 points, x and y per line. File: friedman-maxmin2d/coords/n40.txt.

```
0.325640920663 -0.402881572715 -4.218090688291 -2.754409301140
-2.123872706769 0.812120492850 -3.243157401665 -2.531911915476
0.972966577990 -3.024812512005 -3.446686081018 -3.510980900114
-3.622465572627 0.970593708489 -4.509184560535 -0.790953471630
-0.648499922162 -4.149981735502 0.261660823703 -2.321929787849
-0.170048018242 0.465618557307 -4.214406305482 0.164612214416
-0.367864728045 -1.123332761353 0.804885268385 0.688115942971
-2.942310541053 0.237525408107 -0.863553666950 -0.254832631331
-1.658376047346 -2.365838878180 -2.574610328009 -0.849684205033
-2.591310478486 -4.028989180543 1.425015305239 -2.132819267527
1.570046841219 -1.143392234558 -2.387781799132 -3.049920195905
-1.125131817481 0.761954408041 -0.830353562429 1.717520094087
-1.634175803189 -4.318632432394 0.254530840291 -3.720405849232
1.300574207290 -0.180384187051 -1.829094451716 1.767686178896
0.124730236811 1.421184243352 -1.627476220222 -1.261722949024
-0.477009141997 -3.038607305443 -3.534251273908 -0.568456085966
-2.804027738343 1.545188793232 -3.509961159947 -1.568161037621
-4.484894446573 -1.790658423285 -1.430647123835 -3.339563447756
-1.858247785710 -0.151955957081 -2.513751649878 -1.847830597751
-0.704738065508 -2.064882735866 0.598534161166 -1.380379813335
```